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REMOTELY CONTROLLED VALVE OPERATOR

Abstract

A remotely controlled valve operator includes a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion configured to operatively engage a valve cap of a valve; and a drive system comprising a motor and a sprocket, the motor remotely controlled by a controller and configured to drive rotation of the sprocket, the sprocket engaging the toothed gear portion of the cap actuator to rotate the cap actuator between an activated position and a deactivated position.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to water metering. More specifically, this disclosure relates to a remotely controlled valve operator for a meter valve.

BACKGROUND

[0002] Meter assemblies typically comprise a meter and a meter valve. The meter valve controls the flow of fluid from a pipeline system into the meter, and the meter measures a flow rate of the water or other fluid through the meter assembly. Meter assemblies often are contained within a housing comprising a removable lid. To operate the meter valve, a human operator must typically remove the lid from the housing and manually actuate the meter valve.

SUMMARY

[0003] It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended neither to identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.

[0004] Disclosed is a remotely controlled valve operator comprising a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion configured to operatively engage a valve cap of a valve; and a drive system comprising a motor and a sprocket, the motor remotely controlled by a controller and configured to drive rotation of the sprocket, the sprocket engaging the toothed gear portion of the cap actuator to rotate the cap actuator between an activated position and a deactivated position.

[0005] Also disclosed is a remotely controlled valve assembly comprising a valve comprising a valve body and a valve cap; and a remotely controlled valve operator comprising: a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion operatively engaged with the valve cap; and a drive system comprising a motor and a sprocket, the motor remotely controlled by a controller and configured to drive rotation of the sprocket, the sprocket engaging the toothed gear portion of the cap actuator to rotate the cap actuator between an activated position and a deactivated position; wherein, in the activated position, the cap actuator actuates the valve cap to an open position and fluid is permitted to flow through the valve, and in the deactivated position, the cap actuator actuates the valve cap to a closed position and fluid is prohibited from flowing through the valve.

[0006] Additionally, disclosed is a method of remotely controlling a valve, the method comprising providing a remotely controlled valve operator, the remotely controlled valve operator comprising: a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion engaging a valve cap of the valve; and a drive system comprising a sprocket engaging the toothed gear portion of the cap actuator and a motor for driving rotation of the sprocket, the motor remotely controlled by a controller; remotely actuating the motor with the controller to rotate the sprocket; rotating the cap actuator between a deactivated position and an activated position with the sprocket; and actuating the valve cap with the cap actuator between a closed position, wherein fluid is prohibited from flowing through the valve, and an open position, wherein fluid is permitted to flow through the valve.

[0007] Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The features and components of the following figures are illustrated to emphasize the general principles of the present disclosure. Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

[0009] FIG. **1** is a left-side perspective view of a meter assembly comprising a meter, a meter valve, and a remotely controlled valve operator, in accordance with one aspect of the present disclosure.

[0010] FIG. **2** is a perspective view of the meter valve of FIG. **1**.

[0011] FIG. **3** is a bottom view of a cap actuator and a drive system of the remotely controlled valve operator of FIG. **1**.

[0012] FIG. **4** is a left-side view of the cap actuator and the drive system of FIG. **3** engaged with the meter valve of FIG. **1**.

[0013] FIG. **5** is a cross-sectional view taken along line **5-5** in FIG. **4**.

[0014] FIG. **6** is a left-side perspective view of an operator housing of the remotely controlled valve operator of FIG. **1**.

[0015] FIG. **7** is a cross-sectional view taken along line **7-7** in FIG. **6**.

[0016] FIG. **8** is a perspective view of a torque transmission device of the remotely controlled valve operator of FIG. **1**.

[0017] FIG. **9** is a perspective view of the torque transmission device of FIG. **8**.

[0018] FIG. **10** is a perspective view of a foot of the torque transmission device of FIG. **9**.

[0019] FIG. **11** is a left-side perspective view of a valve assembly comprising the meter valve and the remotely controlled valve operator, in accordance with another aspect of the present disclosure.

[0020] FIG. **12** is an exploded view of the valve assembly of FIG. **11**.

DETAILED DESCRIPTION

[0021] The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and the previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

[0022] The following description is provided as an enabling teaching of the present devices, systems, and/or methods in its best, currently known aspect. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the present devices, systems, and/or methods described herein, while still obtaining the beneficial results of the present disclosure. It will also be apparent that some of the desired benefits of the present disclosure can be obtained by selecting some of the features of the present disclosure without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present disclosure are possible and can even be desirable in certain circumstances and are a part of the present disclosure. Thus, the following description is provided as illustrative of the principles of the present disclosure and not in limitation thereof.

[0023] As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an element” can include two or more such elements unless the context indicates otherwise.

[0024] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0025] For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.

[0026] As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

[0027] The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular aspect.

[0028] Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed that while specific reference of each various individual and collective combinations and permutations of these may not be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspect or combination of aspects of the disclosed methods.

[0029] Disclosed is a remotely controlled valve operator and associated methods, systems, devices, and various apparatus. Example aspects of the remotely controlled valve operator can comprise a cap actuator and a drive system. It would be understood by one of skill in the art that the remotely controlled valve operator is described in but a few exemplary embodiments among many. No particular terminology or description should be considered limiting on the disclosure or the scope of any claims issuing therefrom.

[0030] FIG. **1** is a perspective view of a meter assembly **100**, in accordance with one aspect of the present disclosure. In example aspects, the meter assembly **100** can be housed within a meter box **105**, which may be located above or below ground. The meter box **105** may or may not comprise a lid in example aspects. The meter box **105** is illustrated as transparent in the present aspect to allow visibility of the meter assembly **100** housed therein. According to example aspects, the meter assembly **100** can comprise a meter **110**, a meter valve **120**, and a remotely controlled valve operator **130** coupled to the meter valve **120** and configured to control the actuation of the meter valve **120** between a closed configuration and an open configuration. The meter **110** can comprise a meter main case **112** and a meter cap **114**. A water meter register (not shown) can be enclosed by the meter main case **112** and the meter cap **114**. Example aspects of the meter assembly **100** can further comprise a meter swivel nut (not shown) connecting the meter valve **120** to the meter **110**. Example aspects of meter swivel nuts are disclosed in U.S. application Ser. No. 18/386, 197, filed Nov. 1, 2023, which is hereby specifically incorporated by reference herein in its entirety.

[0031] According to example aspects, the meter assembly **100** can be connected to a pipeline system carrying fluid therethrough. The pipeline system can be, for example, a municipal water pipeline system carrying water to commercial or residential buildings. The meter valve **120** can control the flow of fluid from the pipeline system into the meter **110**. The meter valve **120** can be a ball angle meter valve **122** in the present aspect; however, in other aspects, the meter valve **120** can be any other suitable type of valve known in the art. The meter **110** of the meter assembly **100** can measure a flow rate of the water or other fluid through the meter assembly **100**. Examples of such meter valves and meters are disclosed in U.S. Pat. No. 7,854,165, which issued on Dec. 21, 2010, U.S. Pat. No. 9,664,550, which issued on May 30, 2017, and U.S. application Ser. No. 17/683,111, filed on Feb. 28, 2022, each of which is hereby specifically incorporated by reference herein in its entirety. In other aspects, the pipeline system can be any other suitable type of pipeline system, and the meter **110** can measure the flow rate any other suitable fluid, including liquids and gases, therethrough. In other aspects, the valve operator **130** can be used on any other suitable type of valve and/or in any other suitable application outside of metering.

[0032] The valve operator **130** can engage the meter valve **120** and can be remotely controlled to actuate the meter valve **120** between the open configuration and the closed configuration. In the open configuration, fluid can flow through the meter valve **120** into the meter **110**, and in the closed configuration, fluid can be prohibited from flowing into the meter **110**. The meter valve **120** can comprise a valve body **124** and a valve cap **126**. According to example aspects, the valve operator **130** can comprise a cap actuator **132** configured to engage the valve cap **126** of the meter valve **120** and a drive system **136** configured to drive movement of the cap actuator **132** between an activated position and a deactivated position. In the activated position of the cap actuator **132**, the cap actuator **132** can apply a torque to the valve cap **126** to rotate or otherwise orient the valve cap **126** to an open position, relative to the valve body **124**. In the deactivated position of the cap actuator **132**, the cap actuator **132** can apply a torque to the valve cap **126** to rotate or otherwise orient the valve cap **126** in a closed position, relative to the valve body **124**. In the closed position of the valve cap **126**, the meter valve **120** can be arranged in the closed configuration to prohibit fluid flow into the meter **110**. In the open position of the valve cap **126**, the meter valve **120** can be arranged in the open configuration to allow fluid flow in the meter **110**.

[0033] In some aspects, the drive system **136** can include a toothed drive sprocket **138**, which can be driven by a motor **410** (shown in FIG. **4**), for example. The toothed drive sprocket **138** can rotationally engage a toothed gear portion **134** of the cap actuator **132** to rotate the cap actuator **132** between the activated and deactivated positions, as described in further detail below. In other aspects, additional gears may be arranged between the toothed drive sprocket **138** and the toothed gear portion **134** to form a gear train. Example aspects of the valve operator **130** can further comprise an operator housing **142**. According to example aspects of the valve operator **130**, the cap actuator **132** or portions, thereof and/or the drive system **136** or portions thereof can be housed within the operator housing **142**. In example aspects, the cap actuator **132** and/or the operator housing **142** can comprise a plastic material. In other aspects, the cap actuator **132** and/or the operator housing **142** can comprise a metal material, such as, for example, brass, iron, or stainless steel. In some aspects, the cap actuator **132** can comprise die cast aluminum or zinc, for example and without limitation. Moreover, in example aspects, the toothed drive sprocket **138** can comprise a metal material. In example aspects, the toothed drive sprocket **138** can be metal injection molded (MIM) or powder sintered pressed metal. In other aspects, the cap actuator **132**, the operator housing **142**, and/or the toothed drive sprocket **138** can comprise any other suitable material and/or can be manufactured by any suitable method.

[0034] The valve operator **130** of the present aspect can further comprise a torque transmission device **150**. Other aspects of the valve operator **130** may not comprise the torque transmission device **150**, such as the aspect disclosed in FIGS. **11** and **12**. In example aspects, rotating the valve cap **126** of the meter valve **120** can require a significant amount of torque. The drive system **136** of the valve operator **130** can provide a suitable amount of torque to the valve cap **126** to rotate the valve cap **126** between the open and closed positions, and the torque can be transferred to opposing box sidewalls **106***a*,*b* of the meter box **105** by the torque transmission device **150**. The torque transmission device **150** can also aid in holding the cap actuator **132** down on the valve cap **126**. In the present aspect, the torque transmission device **150** can comprise a torque bar **152** and opposing feet **154***a*,*b* arranged at opposing bar ends of the torque bar **152**. The opposing feet **154***a*,*b* can be secured to the opposing box sidewalls **106***a,b* of the meter box **105** by any suitable fastener, including but not limited to, adhesive, such as glue or double-sided tape, screws, nails, sharp teeth configured to grip the opposing box sidewalls **106***a,b*, or the like.

[0035] According to example aspects, the torque transmission device **150** can be adjustable in length to accommodate meter boxes **105** of varying dimensions. Typical meter boxes **105** all comprise substantially flat, opposing box sidewalls **106***a,b*, regardless of the meter box **105** dimensions, and the feet of the torque transmission device **150** can engage with any suitable flat, opposing box sidewalls **106***a,b*. Providing the length adjustability feature of the torque transmission device **150** can allow the valve operator **130** to be used with varying sizes of meter boxes **105**. In other aspects, however, the length of the torque transmission device **150** may not be adjustable. Additionally, torquing against the opposing box sidewalls **106***a,b* of the meter box **105** (as opposed to, for example, against the valve body **124** of the meter valve **120**) allows for the valve operator **130** to interface only with the valve cap **126**. Because the geometry of valve caps **126** are controlled by industry standards, and therefore are typically the same in size and shape across different types of valves, the valve operator **130** can be used with a variety of shapes and sizes of meter valves **120**.

[0036] FIG. **2** illustrates the meter valve **120** according to an example aspect of the present disclosure. In the present aspect, the meter valve **120** can be the ball angle meter valve **122**; however, in other aspects, the meter valve **120** can be any other suitable type of valve known in the art. The meter valve **120** can define the valve body **124** and valve cap **126**. As shown, the valve cap **126** can be rotationally mounted to the valve body **124** at a top body end **222** of the valve body **124**. The valve cap **126** can define a cap flange **224** extending upward from a cap base **226**. The cap flange **224** can be substantially rectangular in the present aspect. Additionally, according to example aspects, the cap flange **224** can define a plurality of flange sides **228**, and each of the flange sides **228** can taper upward and away from the cap base **226**, relative to the orientation shown. The flange sides **228** of the cap flange **224** may be tapered because the valve cap **126** may be formed by casting and require a suitable draft for molding. The tapered flange sides **228** can cause a rising or separating force as torque is applied by the valve operator **130** (shown in FIG. **1**) to rotate the valve cap **126** between the open and closed positions. However, the torque transmission device **150** (shown in FIG. **1**) can securely attach the valve operator **130** to the valve cap **126** to prevent the cap actuator **132** (shown in FIG. **1**) from rising off of the valve cap **126** as the torque is applied. In some aspects, a through-hole **240** can be defined through valve cap **126**, as shown. In the present aspect, the through-hole **240** can be defined through the cap flange **224** of the valve cap **126**.

[0037] The valve body **124** can define a valve inlet end **230** and a valve outlet end **232** opposite the valve inlet end **230**. A fluid pathway **510** (shown in FIG. **5**) can extend between the valve inlet end **230** and the valve outlet end **232**. A ball can be disposed within the fluid pathway **510** of the valve body **124** between the valve inlet end **230** and the valve outlet end **232** and can operatively connected to the valve cap **126**. The valve cap **126** can be selectively rotated between the open and closed positions by the valve operator **130** (shown in FIG. **1**) to allow or prevent fluid flow therethrough, respectively. In the open position of the valve cap **126**, an opening formed through the ball can be aligned with the fluid pathway **510** of the valve body **124** to allow fluid flow therethrough, and in the closed position of the valve cap **126**, the opening of the ball can be blocked to prohibit fluid flow therethrough. Various component of the meter valve **120**, including the valve cap **126**, can be formed from a metal material, such as brass for example and without limitation. In other aspects, the meter valve **120** can comprise any other suitable material or combination of materials, including various metals and plastics.

[0038] FIG. **3** illustrates a bottom view of the cap actuator **132** and the toothed drive sprocket **138** of the valve operator **130** (shown in FIG. **1**). According to example aspects, the cap actuator **132** can comprise a cap engagement portion **310** configured to engage the valve cap **126** (shown in

FIGS. **1** and **2**) and the toothed gear portion **134** extending outward from the cap engagement portion **310**. The toothed gear portion **134** can be formed monolithically with the cap engagement portion **310** in the present aspect (i.e., formed a singular component that constitutes a single material without joints or seams). In other aspects, the toothed gear portion **134** and the cap engagement portion **310** can be formed separately and coupled together by any suitable fastener, including but not limited to, welding, screws, adhesives, etc.

[0039] The toothed gear portion **134** can be substantially arcuate and can extend at least partially about a circumference of the cap engagement portion **310**, as shown. The toothed gear portion **134** can define a plurality of outer gear teeth **312**, distal to the cap engagement portion **310**. A gear tooth gap **314** can be defined between each adjacent pair of outer gear teeth **312**. The toothed drive sprocket **138** can be substantially annular and can define a plurality of outer sprocket teeth **316** about a circumference of the toothed drive sprocket **138**. A sprocket tooth gap **318** can be defined between each adjacent pair of outer sprocket teeth **316**. Each of the outer sprocket teeth **316** can engage a corresponding one of the gear tooth gaps **314** and each of the outer gear teeth **312** can engage a corresponding one of the sprocket tooth gaps **318** as the sprocket is rotated by the motor **410** (shown in FIG. **4**) to drive the rotation of the cap actuator **132**.

[0040] In the present aspect, the cap engagement portion **310** can be formed as a cap cover **320**, which can cover and receive at least a portion of the valve cap **126** (shown in FIGS. **1** and **2**). An inner engagement surface **322** of the cap engagement portion **310** can define a cap cavity **324** of the cap actuator **132**. The cap cavity **324** can extend into the cap engagement portion **310** at a bottom actuator end **325** of the cap engagement portion **310**. The cap cavity **324** of the cap actuator **132** can be configured to receive the valve cap **126**, or portions thereof, therein. In the present aspect, the cap cavity **324** can define a base cavity portion **326** configured to receive at least a portion of the cap base **226** (shown in FIG. **2**) of the valve cap **126** and a flange cavity portion **328** configured to receive the cap flange **224** (shown in FIG. **2**) of the valve cap **126**. In other aspects, the cap cavity **324** may receive the cap flange **224**, or a portion thereof, only.

[0041] In some aspects, the cap engagement portion **310** can be secured to the valve cap **126** (shown in FIGS. **1** and **2**) by any suitable fastener, including but not limited to, adhesives, screws, bolts, etc. In some aspects, an intermediate material can be arranged between the valve cap **126** and the inner engagement surface **322** of the cap engagement portion **310**. For example and without limitation, the intermediate material can be an initially adjustable, but ultimately hard material, capable of transmitting the torque between the valve operator **130** (shown in FIG. **1**) and valve cap **126**, such as a two-part liquid epoxy, high density spray foam, or two-part clay epoxy. In some aspects, the two-part liquid epoxy or the high density foam spray can be housed inside of a bag that can fill and conform to the space between the valve cap **126** and the inner engagement surface **322** and can then harden. In some aspects, the two-part clay epoxy can be mixed by hand and pressed into the cap cavity **324** of the cap engagement portion **310**, and then can be pressed against the valve cap **126** to conform thereto. The cap cavity **324** can thereby accommodate valve caps **126** of varying shapes and/or sizes. In other aspects, the cap cavity **324** can be sized and shaped to receive the valve cap **126**, or portions thereof, with minimal clearance therebetween.

[0042] FIG. **4** illustrates the cap engagement portion **310** of the cap actuator **132** coupled to the valve cap **126** of the meter valve **120**. FIG. **4** also illustrates the drive system **136** operatively engaged with the toothed gear portion **134** of the cap actuator **132**. The cap base **226** (shown in FIG. **2**) and the cap flange **224** (shown in FIG. **2**) of the valve cap **126** can be substantially received within the cap cavity **324** (shown in FIG. **3**) of the cap engagement portion **310**. The toothed gear portion **134** of the cap actuator **132** can extend substantially radially outward from the cap engagement portion **310**, relative to a cavity axis **420** extending through the cap cavity **324** of the cap actuator **132**. The toothed gear portion **134** can extend from an outer actuator sidewall **422** of the cap actuator **132**, proximate to the bottom actuator end **325** of the cap actuator **132**, opposite a top actuator end **426** thereof. In other aspects, the toothed gear portion **134** can be arranged elsewhere along the cap actuator **132** between the top actuator end **426** and the bottom actuator end **325**. The outer gear teeth **312** of the toothed gear portion **134** can matingly engage the toothed drive sprocket **138**, and the outer sprocket teeth **316** of the toothed drive sprocket **138** can matingly engage the toothed gear portion **134**, as previously described. In some aspects, a fastener hole, such as a pin hole **424**, can extend laterally through the outer actuator sidewall **422** of the cap engagement portion **310**. The pin hole **424** can be aligned with the through-hole **240** (shown in FIG. **2**), and a pin **428** or other suitable fastener can extend through each of the pin hole **424** and the through-hole **240** to pin the cap engagement portion **310** to the valve cap **126**.

[0043] Example aspects of the drive system **136** can comprise at least the toothed drive sprocket **138** and the motor **410** for driving the rotation of the toothed drive sprocket **138**. According to example aspects, a substantially annular sprocket flange **430** can extend axially from a center of the substantially annular toothed drive sprocket **138**, relative to a sprocket axis **432** extending through the center of the toothed drive sprocket **138**. As shown, a drive stem **440** can extend from and be rotationally driven by the motor **410**. The drive stem **440** can engage a stem opening **520** (shown in FIG. **5**) extending axially into the sprocket flange **430**, relative to the sprocket axis **432**, to operatively couple the motor **410** to the toothed drive sprocket **138**. In some aspects, each of the drive stem **440** and the stem opening **520** of the sprocket flange **430** can define a cross-sectional shape of a truncated circle, for example and without limitation. In other aspects, the cross-sectional shape of the drive stem **440** and stem opening **520** can be rectangular, triangular, semi-circular, or any other suitable shape.

[0044] The drive stem **440** can be rotationally driven by the motor **410** about the sprocket axis **432**, which in turn can rotate the toothed drive sprocket **138** about the sprocket axis **432**. In example aspects, power can be provided to the motor **410** by a power source, such as batteries **1210** (shown in FIG. **12**) or solar power, for example and without limitation. The engagement of the toothed drive sprocket **138** with the toothed gear portion **134** of the cap actuator **132** can rotate the cap actuator **132** about the cavity axis **420** of the cap actuator **132** as the toothed drive sprocket **138** is rotated. The cap engagement portion **310** of the cap actuator **132** can be mounted to the valve cap **126** of the meter valve **120**, and the valve cap **126** can be rotated between the open and closed positions as the cap actuator **132** is rotated. In example aspects, the motor **410** can be selectively driven in both a forward direction and a reverse direction to allow the valve cap **126** to be rotated in a first direction to the open position and an opposite second direction to the closed position. In example aspects, the valve cap **126** can be rotated about 90° in the first direction to the open position and about 90° in the second direction to the closed position. In example aspects, the valve cap **126** can further be rotated to any number of intermediate positions between the open and closed positions to selectively increase or decrease the flow of fluid through the meter valve **120**.

[0045] In example aspects, a motor plate **450** can be mounted to a bottom motor end **412** of the motor **410**. The drive stem **440** can extend through a plate opening **530** (shown in FIG. **5**) formed through the motor plate **450**. The motor plate **450** can transmit toque from the motor **410** to the operator housing **142** (shown in FIG. **1**), as described in further detail below. Moreover, in the present aspect, a packing **460**, such as an O-ring **462**, can be mounted to the sprocket flange **430**. The O-ring **462** can interface with the operator housing **142** to create a fluid-tight seal between the operator housing **142** and the sprocket flange **430**, to protect the motor **410** housed within the operator housing **142** from moisture and other undesirable elements.

[0046] FIG. **5** illustrates a cross-sectional view of the cap actuator **132** and the drive system **136** of valve operator **130** (shown in FIG. **1**) mounted to the meter valve **120**, taken along line **5-5** in FIG. **4**. As shown, the cap engagement portion **310** of the cap actuator **132** can engage the valve cap **126** of the meter valve **120**. More specifically, in the present aspect, the cap base **226** of the valve cap **126** of the meter valve **120** can be at least partially received within the base cavity portion **326** of the cap cavity **324**, and the cap flange **224** can be received within the flange cavity portion **328** of the cap cavity **324**. As previously described, the drive system **136** can actuate the cap actuator **132**,

which in turn can drive the rotation of the valve cap **126** between the open and closed positions. More specifically, the toothed drive sprocket **138** of the drive system **136** can engage the toothed gear portion **134** of the cap actuator **132** as previously described. The drive stem **440** can be operatively connected to the toothed drive sprocket **138** and can be rotationally driven by the motor **410** of the drive system **136**. Rotating the drive stem **440** can thereby rotate the toothed drive sprocket **138**, which can in turn rotate the cap actuator **132**, which can in turn rotate the valve cap **126** of the meter valve **120** between the open and closed positions.
[0047] The motor **410** and drive stem **440** are illustrated as transparent in the present aspect to allow visibility of the motor plate **450** mounted to the motor **410** at the bottom motor end **412** of the motor **410**. As shown, the motor plate **450** can define a plurality of plate mounting holes **532**, each of which can be aligned with a corresponding motor mounting hole **534** formed in the motor **410**. A fastener, such a screw or bolt for example and without limitation, can engage each corresponding pair of plate mounting holes **532** and motor mounting holes **534** to couple the motor plate **450** to the motor **410**.
[0048] FIGS. **6** and **7** illustrate a remotely controlled valve assembly **600** comprising the meter valve **120** and the valve operator **130**. FIG. **6** illustrates the full valve operator **130** mounted to the valve cap **126** of the meter valve **120**, and FIG. **7** illustrates a cross-sectional view taken along line **7-7** in FIG. **6**. As shown, the valve operator **130** comprises the operator housing **142**, which can at least partially house the drive system **136** and the cap actuator **132**. For example, in the present aspect, the operator housing **142** can comprise an actuator housing portion **610** for at least partially housing the cap actuator **132** and a drive housing portion **620** for at least partially housing the drive system **136**. As shown, the actuator housing portion **610** can define an actuator housing cavity **712** (shown in FIG. **7**) that can house a portion of the cap engagement portion **310** of the cap actuator **132**, including the cap flange **224** (shown in FIG. **7**) and at least a portion of the cap base **226** (shown in FIG. **7**). However, the toothed gear portion **134**, including the outer gear teeth **312**, and the bottom actuator end **325** of the cap engagement portion **310** can be arranged external to the actuator housing portion **610**.
[0049] The drive housing portion **620** of the operator housing **142** can define a drive housing cavity **714** (shown in FIG. **7**) that can at least partially house the drive system **136**. More specifically, in the present aspect, the drive housing portion **620** can house the motor **410** (shown in FIG. **7**), the motor plate **450** (shown in FIG. **7**), the drive stem **440** (shown in FIG. **7**), and a portion of the toothed drive sprocket **138**. Specifically, the sprocket flange **430** (shown in FIG. **7**) of the toothed drive sprocket **138** can extend into the drive housing portion **620** through a housing opening **710** (shown in FIG. **7**) formed at a bottom housing end **624** of the drive housing portion **620**. The O-ring **462** (shown in FIG. **7**) mounted on the sprocket flange **430** can be arranged within the housing opening **710** to form a fluid-tight seal between the sprocket flange **430** and the drive housing portion **620**, thereby preventing moisture and other undesirable elements from entering the drive housing cavity **714**. However, the outer sprocket teeth **316** (shown in FIG. **3**) of the toothed drive sprocket **138** can be arranged external to the drive housing portion **620**, as shown. The outer sprocket teeth **316** of the toothed drive sprocket **138** can thereby engage the outer gear teeth **312** of the toothed gear portion **134** external to the operator housing **142**. In other aspects, however, the toothed drive sprocket **138** and/or the toothed gear portion **134** may be housed within the operator housing **142**. In the cross-sectional view of FIG. **7**, the interior of the motor **410** is illustrated as solid for the sake of simplicity; however, various electronic components can be housed therein for driving the motor **410**.
[0050] In some aspects, the operator housing **142** can comprise multiple housing segments **630** that can be formed separately and assembled together. In other aspects, the operator housing **142** can be monolithically formed. In the present aspect, the operator can comprise two of the housing segments **630**. A first housing segment **632** of the two housing segments **630** can comprise the actuator housing portion **610** and a first portion **634** of the drive housing portion **620**. A second

housing segment **636** can comprise a second portion **638** of the drive housing portion **620**. The first housing segment **632** can be joined to the second housing segment **636** as a seam **660**, as shown. The first housing segment **632** can be coupled to the second housing segment **636** by any suitable fasteners or fastening techniques, including welding, adhesives, bolts, screws, and the like.

[0051] Additionally, the operator housing **142** can define a mounting bracket **640** for mounting the torque transmission device **150** (shown in FIG. **1**) to the valve operator **130**. In example aspects, the mounting bracket **640** can extend outward from an outer housing surface **650** of the drive housing portion **620**. In other aspects, the mounting bracket **640** can be positioned elsewhere on the operator housing **142**. The mounting bracket **640** can define a bar opening **642** extending therethrough from a first bracket side **644** of the mounting bracket **640** to a second bracket side **646** of the mounting bracket **640**. The torque bar **152** (shown in FIG. **1**) of the torque transmission device **150** can be configured to extend through the bar opening **642** of the mounting bracket **640**, as illustrated in FIG. **1**.

[0052] The actuation of the valve operator **130** can be remotely controlled. Example aspects of the valve operator **130** can comprise a controller for operating the motor **410** (shown in FIG. **7**) of the drive system **136** and for selectively changing the direction of the motor **410** between the forward and reverse directions. The controller can be completely wireless or can be wired to various components of the valve operator **130**, such as the motor **410**. The controller can allow an external operator, such as a human operator or a computer, to remotely control the operation of the meter valve **120**, i.e., to remotely control the selective orientation of the valve cap **126** in the closed, open, or intermediate positions by actuating the valve operator **130**. The controller can be remotely controlled from a remote operation device, such as a mobile phone, tablet, computer, or the like, which can be wirelessly connected to the controller. In example aspects, a program or app can be downloaded onto the remote operation device, through which the operator can wirelessly send control signals to the controller, and the controller can actuate the valve operator **130** in response to the control signals. Thus, the meter valve **120** does not need to be physically accessed by a human operator in order to be selectively configured in the open or closed configuration. In some aspects, the controller can be a Bluetooth® controller. In other aspects, any other suitable wireless communication technique(s) may be implemented for remotely controlling the valve operator **130** with the remote operation device and controller.

[0053] In some example aspects, the valve operator **130** can further comprise any or all of a time controller and/or microprocessor, integral communications (e.g., cellular node, radio, short wave, LPWAN), and/or a wifi or Bluetooth antenna. In some example aspects, the valve operator **130** can further comprise end limit switches, optical isolators, indicator lights, and or physical indicators for visually indicating the position of the valve cap **126** (e.g., open, closed, or any number of intermediate positions). For example and without limitation, such a physical indicator could include a molded plastic “ON” that appears in a window when the meter valve **120** is in the fully open configuration and a molded plastic “OFF” that appears in the window when the meter valve **120** is in the fully closed configuration.

[0054] FIGS. **8** and **9** illustrate perspective views of the torque transmission device **150**. The torque transmission device **150** can comprise the torque bar **152** and the two opposing feet **154***a,b* mounted to the torque bar **152**. In example aspects, the torque bar **152** can comprise an elongate threaded rod **810**. The threaded rod **810** can define a first rod end **812** and a second rod end **914** (shown in FIG. **9**) opposite the first opposing end. A first foot **154***a* of the two opposing feet **154***a,b* can be mounted to the threaded rod **810** proximate to the first rod end **812**, and the second foot **154***b* of the two opposing feet **154***a,b* can be mounted to the threaded rod **810** proximate to the second rod end **914**. The threaded rod **810** can define a central rod section **816** between the first foot **154***a* and the second foot **154***b*, and the central rod section **816** can extend through the bar opening **642** (shown in FIG. **6**) of the mounting bracket **640** (shown in FIG. **6**) of the valve operator **130** (shown in FIG. **1**).

[0055] In example aspects, a first positioning nut **820** and a second positioning nut **822** can be mounted to the central rod section **816** of the threaded rod **810**. Each of the first positioning nut **820** and the second positioning nut **822** can be hex nuts in the present aspect; however, in other aspects, the first and second positioning nuts **820**,**822** can be any other suitable type of nut. The first positioning nut **820** can be tightened against the first bracket side **644** (shown in FIG. **6**) of the mounting bracket **640** (shown in FIG. **6**) and the second positioning nut **822** can be tightened against the second bracket side **646** (shown in FIG. **6**) of the mounting bracket **640** to securely couple the threaded rod **810** to the mounting bracket **640** at a desired position along the central rod section **816**. Prior to tightening the first and second positioning nuts **820**,**822**, the central rod section **816** of the threaded rod **810** can be slid back and forth laterally within the bar opening **642** (shown in FIG. **6**) to position the mounting bracket **640** at the desired position along the central rod section **816**.

[0056] According to example aspects, each of the first foot **154***a* and the second foot **154***b* can define a rod fastener **830**, a body portion **832**, and a wall bracket **838**. The rod fastener **830** can be positioned at a proximal end **834** of the body portion **832**, and the wall bracket **838** can be positioned at a distal end **836** of the body portion **832**. In the present aspect, the body portion **832** can define the shape of a truncated square pyramid. The proximal end **834** of the body portion **832** can be the truncated end, as shown. In other aspects, the first foot **154***a* and/or the second foot **154***b* can define any other suitable shape.

[0057] The rod fastener **830** can be arranged adjacent to the central rod section **816** and can couple the corresponding first foot **154***a* or second foot **154***b* to the threaded rod **810**. The rod fastener **830** can define a threaded bore **1010** (shown in FIG. **10**), and the threaded rod **810** can rotationally engage the threaded bore **1010**. As shown, in example aspects, the first rod end **812** and the second rod end **914** (shown in FIG. **9**) can extend through the threaded bore **1010** of the corresponding rod fastener **830** and into a foot cavity **850** defined by the corresponding body portion **832**. The first foot **154***a* and/or second foot **154***b* can be rotated on the threaded rod **810** towards or away from the central rod section **816** to decrease or increase the length of the torque transmission device **150**, respectively. The length of the torque transmission device **150** can thereby be selectively adjusted to accommodate meter boxes **105** (shown in FIG. **1**) of varying dimensions.

[0058] The wall bracket **838** can be formed as a substantially planar plate **840** at the distal end **836** of the corresponding first foot **154***a* or second foot **154***b*. The wall bracket **838** can be substantially square shaped in the present aspect, and can define a central cavity opening **842** allowing access to the foot cavity **850**. In other aspects, the wall bracket **838** can define any other suitable shape and/or may not define the central cavity opening **842**. Example aspects of the wall bracket **838** can comprise one or more bracket fastener openings **844**. The wall bracket **838** of the opposing feet **154***a,b* can be arranged to lie substantially flat against the opposing box sidewalls **106***a,b* (shown in FIG. **1**) of the meter box **105**. A bracket fastener can extend through each of the bracket fastener openings **844** and can engage the corresponding box sidewall **106***a,b* to attach the opposing feet **154***a,b* to the opposing box sidewalls **106***a,b*. Prior to attaching the opposing feet **154***a,b* to the opposing box sidewalls **106***a,b*, the opposing feet **154***a,b* can be readjusted as needed along the length of the threaded rod **810**, as previously described, to properly abut the corresponding wall bracket **838** against the corresponding box sidewall **106***a,b*.

[0059] FIG. **10** illustrates a perspective view of the one of the opposing feet **154** of the torque transmission device **150** (shown in FIG. **1**). For example, the first foot **154***a* is shown in the present view, which can be substantially the same as the second foot **154***b* (shown in FIG. **1**). As shown, the rod fastener **830** can be arranged at the proximal end **834** of the body portion **832** of the first foot **154***a*, and the wall bracket **838** can be arranged at the distal end **836** of the body portion **832**. In the present aspect, the rod fastener **830** can comprise a tube portion **1012** formed monolithically with the body portion **832** and the wall bracket **838**. The rod fastener **830** can further comprise a fastener nut **1014** mounted within the tube portion **1012**. In example aspects, the tube portion **1012** can be substantially hexagonal and the fastener nut **1014** can be a hex nut, similar to or the same as the first and second positioning nuts **820**,**822** (shown in FIG. **8**). In other aspects, the fastener nut **1014** can be any other suitable type of nut. The fastener nut **1014** can be retained within the tube portion **1012** of the rod fastener **830** by friction fit or by any suitable fastening device or technique, including adhesives, welding, rivets, and the like. As shown, the fastener nut **1014** can define the threaded bore **1010** configured to rotationally receive to the threaded rod **810** (shown in FIG. **8**) therethrough.

[0060] FIGS. **11** and **12** illustrate perspective and exploded views respectively of the valve assembly **600** according to another example aspect of the present disclosure. The valve assembly **600** can comprise the meter valve **120** and the valve operator **130**. The meter valve **120** can be substantially the same as the meter valve **120** of the previous aspect. However, various features of the valve operator **130** can differ from valve operator **130** of the previous aspect, as described in further detail below. The valve operator **130** of the present aspect can be mounted to the valve cap **126** of the meter valve **120** and can be remotely controlled to actuate the valve cap **126** between the open and closed positions, and to thereby actuate the meter valve **120** between the open and closed configurations, respectively. Similar to the valve operator **130** of FIGS. **1-10**, the valve operator **130** of the present aspect can comprise the cap actuator **132** configured to actuate the valve cap **126** of the meter valve **120** and the drive system **136** (shown in FIG. **12**) configured to drive the cap actuator **132** between the activated and deactivated positions.

[0061] The cap actuator **132** can comprise the cap engagement portion **310** and the toothed gear portion **134**. The drive system **136** can comprise the motor **410** (shown in FIG. **12**), the drive stem **440** (shown in FIG. **12**), and the toothed drive sprocket **138** (shown in FIG. **12**). The valve operator **130** can comprise the operator housing **142**, which in the present aspect does not include the actuator housing portion **610** (shown in FIG. **6**). Rather, the operator housing **142** of the present aspect can comprise the drive housing portion **620** and can further comprise a housing lid **1110**. The housing lid **1110** can couple the drive housing portion **620** of the operator housing **142** to the cap actuator **132**, and further can prevent the cap actuator **132** from rising off of the valve cap **126** as the torque is applied, as described in further detail below. The motor **410** and at least a portion of the drive stem **440** can be housed within the drive housing portion **620**. In some aspects, the toothed drive sprocket **138** can comprise the sprocket flange **430** (shown in FIG. **4**), and the drive stem **440** can engage the sprocket flange **430** as previously described. The sprocket flange **430** can extend into the drive housing cavity **714** (shown in FIG. **12**) of the drive housing portion **620** through the housing opening **710** (shown in FIG. **12**). The housing opening **710** can be formed through an inner housing shelf **1112** of the drive housing portion **620**. The inner housing shelf **1112** can be positioned proximate to a top housing end **1116** of the drive housing portion **620**. The top housing end **1116** of the drive housing portion **620** can be arranged opposite the bottom housing end **624** thereof.

[0062] The power source (e.g., the batteries **1210**, shown in FIG. **12**) of the drive system **136** (shown in FIG. **12**) can be housed within either the housing lid **1110** of the operator housing **142** or the drive housing portion **620** of the operator housing **142**. In the present aspect, the drive housing portion **620** can comprise an outer housing shelf **1114** configured to extend over top of the toothed drive sprocket **138** (shown in FIG. **12**) at the top housing end **1116**. The outer housing shelf **1114** can be configured to support the housing lid **1110** thereon. In the present aspect, the toothed drive sprocket **138** can be arranged external to the drive housing cavity **714** (shown in FIG. **12**) of the drive housing portion **620** and positioned laterally between the outer housing shelf **1114** and the inner housing shelf **1112**. In example aspects, the outer housing shelf **1114** can define an arcuate indentation **1220** (shown in FIG. **12**) that can be configured to accommodate a portion of the outer actuator sidewall **422** of the cap actuator **132**.

[0063] According to example aspects, the drive housing portion **620** can comprise the two housing segments **630**. The first housing segment **632** can be joined to the second housing segment **636** at the seam **660**, as shown. The first housing segment **632** can be coupled to the second housing segment **636** by any suitable fasteners or fastening techniques, including welding, adhesives, bolts, screws, and the like. In example aspects, the first housing segment **632** can comprise the first portion **634** of the drive housing portion **620**, which can include at least a portion of the inner housing shelf **1112** in some aspects. The first housing segment **632** can further comprise a body engagement portion **1120** configured to engage the valve body **124** of the meter valve **120** at a first valve side **1130** thereof. The body engagement portion **1120** can define a concave pocket **1122** that can fit snugly against the first valve side **1130**, as shown. In some aspects, the body engagement portion **1120** (or the entire first housing segment **632**) can be interchangeable with other body engagement portions **1120** of varying shapes to accommodate different types and sizes of meter valves **120**. The body engagement portion **1120** is described in further detail below. The second housing segment **636** can comprise the second portion **638** of the drive housing portion **620**, which can include at least a portion of the outer housing shelf **1114** in some aspects.

[0064] The valve cap **126** of the meter valve **120** can be actuated between the open and closed positions as previously described. That is, the drive stem **440** (shown in FIG. **12**) can be rotationally driven by the motor **410** (shown in FIG. **12**) about the sprocket axis **432** (shown in FIG. **12**), which in turn can rotate the toothed drive sprocket **138** (shown in FIG. **12**) about the sprocket axis **432**. Power can be provided to the motor **410** by the power source, such as the batteries **1210**. The engagement of the toothed drive sprocket **138** with the toothed gear portion **134** of the cap actuator **132** can rotate the cap actuator **132** about the cavity axis **420** (shown in FIG. **12**) of the cap actuator **132** as the toothed drive sprocket **138** is rotated. The cap engagement portion **310** of the cap actuator **132** can be mounted to the valve cap **126** of the meter valve **120**, and the valve cap **126** can be rotated between the open and closed positions as the cap actuator **132** is rotated.

[0065] In the present aspect, the valve operator **130** does not comprise the torque transmission device **150** previously described. Rather, the body engagement portion **1120** of the operator housing **142** can engage the first valve side **1130** of the valve body **124** of the meter valve **120**, as shown in FIG. **11**. In some aspects, the body engagement portion **1120** can be secured in place against the first valve side **1130** of the valve body **124** by a fastening device of the valve operator **130**, such as, for example and without limitation, a strap, bolt, U-clamp, hose clamp, or the like. In some aspects, the body engagement portion **1120** can extend around the valve body **124** and can define an opposing concave pocket (opposite the concave pocket **1122**) configured to snugly fit against a second valve side **1132** (opposite the first valve side **1130**) of the valve body **124**. The fastening device can encircle the valve body **124** to secure the operator housing **142** thereto. In other aspects, however, the fastening device can comprise any other suitable fasteners or fastening techniques known in the art, including but not limited to, welding, adhesives, bolts, screws, and the like.

[0066] As previously described, example aspects of the cap flange **224** (shown in FIG. **12**) of the valve cap **126** can be tapered on each of the flange sides **228** (shown in FIG. **12**). The flange sides **228** may be tapered because the valve cap **126** maybe formed by casting and require draft for molding. The tapered flange sides **228** can cause a rising or separating force as torque is applied by the valve operator **130** to rotate the valve cap **126** between the open and closed positions. However, the secure attachment of the body engagement portion **1120** to the valve body **124** can prevent the cap engagement portion **310** of the cap actuator **132** from rising off of the valve cap **126** as the torque is applied.

[0067] In example aspects, a method of assembling the valve assembly **600** of FIGS. **11** and **12** can comprise placing the cap engagement portion **310** onto the valve cap **126** with the cap flange **224** of the valve cap **126** engaging the flange cavity portion **328** (shown in FIG. **3**) of the cap engagement portion **310**. In the present aspect, the valve cap **126** may not be pinned to the cap engagement portion **310**. The drive housing portion **620** and motor **410** can then be slid towards the meter valve **120** from the side and the clamp can be tightened. The cap actuator **132** can be prevented from rising off the valve cap **126** because the toothed gear portion **134** of the cap actuator **132** can be constrained between the inner housing shelf **1112** and the outer housing shelf **1114**. The housing lid **1110** can then be glued, or otherwise affixed, to the top housing end **1116** of the drive housing portion **620**.

[0068] One should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

[0069] It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

Claims

1. A remotely controlled valve operator comprising: a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion configured to operatively engage a valve cap of a valve; and a drive system comprising a motor and a sprocket, the motor remotely controlled by a controller and configured to drive rotation of the sprocket, the sprocket engaging the toothed gear portion of the cap actuator to rotate the cap actuator between an activated position and a deactivated position.

2. The remotely controlled valve operator of claim 1, wherein: the cap engagement portion defines a cap cavity extending into the cap engagement portion at a bottom actuator end of the cap actuator; the cap engagement portion define a fastener hole; the valve cap defines a through-hole aligned with the fastener hole; and a fastener extends through each of the fastener hole and the through-hole to couple the cap engagement portion to the valve cap.

3. The remotely controlled valve operator of claim 2, wherein the cap cavity defines a cavity axis extending centrally therethrough, and wherein the toothed gear portion of the cap actuator extends radially outward from the cap engagement portion proximate to the bottom actuator end of the cap actuator.

4. The remotely controlled valve operator of claim 1, further comprising an operator housing, wherein the motor of the drive system is housed within the operator housing, and wherein the sprocket and the toothed gear portion of the cap actuator are arranged external to the operator housing.

5. The remotely controlled valve operator of claim 4, wherein: the drive system further comprises a drive stem extending from and rotationally driven by the motor; the drive stem engages a sprocket flange of the sprocket; and the sprocket flange engages a housing opening of the operator housing.
6. The remotely controlled valve operator of claim 5, wherein a packing is mounted to the sprocket flange and is arranged within the housing opening to form a fluid-tight seal between the sprocket flange and the operator housing.
7. The remotely controlled valve operator of claim 4, wherein: the operator housing comprises a drive housing portion defining a drive housing cavity and an actuator housing portion defining an actuator housing cavity; the motor is housed within the drive housing cavity; and at least a portion of the cap engagement portion of the cap actuator is housed within the actuator housing cavity.
8. The remotely controlled valve operator of claim 1, wherein: the remotely controlled valve operator further comprises a torque transmission device; the torque transmission device comprising a torque bar, a first foot mounted on the torque bar proximate to a first end of the torque bar, and a second foot mounted on the torque bar proximate to an opposite second end of the torque bar; and the first foot is configured to secure to a first wall of a meter box and the second foot is configured to secure to an opposing second wall of the meter box.
9. The remotely controlled valve operator of claim 8, wherein: an operator housing of the remotely controlled valve operator comprises a mounting bracket defining a bar opening therethrough; the torque bar is formed as a threaded rod and extends through the bar opening; a first positioning nut is mounted on the threaded rod and tightened against a first bracket side of the mounting bracket; and a second positioning nut is mounted on the threaded rod and tightened against an opposite second bracket side of the mounting bracket.
10. The remotely controlled valve operator of claim 8, wherein: the torque bar is formed as a threaded rod; each of the first foot and the second foot defines a rod fastener, a wall bracket, and a body portion extending between the rod fastener and the wall bracket; the wall bracket of the first foot is configured to secure to the first wall of the meter box and the wall bracket of the second foot is configured to secure to the opposing second wall of the meter box; the rod fastener of each of the first foot and the second foot defines a threaded bore; and the threaded rod is rotationally engaged with the threaded bore.
11. A remotely controlled valve assembly comprising: a valve comprising a valve body and a valve cap; and a remotely controlled valve operator comprising: a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion operatively engaged with the valve cap; and a drive system comprising a motor and a sprocket, the motor remotely controlled by a controller and configured to drive rotation of the sprocket, the sprocket engaging the toothed gear portion of the cap actuator to rotate the cap actuator between an activated position and a deactivated position; wherein, in the activated position, the cap actuator actuates the valve cap to an open position and fluid is permitted to flow through the valve, and in the deactivated position, the cap actuator actuates the valve cap to a closed position and fluid is prohibited from flowing through the valve.
12. The remotely controlled valve assembly of claim 11, wherein: the cap engagement portion defines a cap cavity extending into the cap engagement portion at a bottom actuator end of the cap actuator; the cap engagement portion defines a fastener hole; the valve cap defines a through-hole aligned with the fastener hole; and a fastener extends through each of the fastener hole and the through-hole to couple the cap engagement portion to the valve cap.
13. The remotely controlled valve assembly of claim 12, wherein the cap cavity defines a cavity axis extending centrally therethrough, and wherein the toothed gear portion of the cap actuator extends radially outward from the cap engagement portion proximate to the bottom actuator end of the cap actuator.
14. The remotely controlled valve assembly of claim 11, further comprising an operator housing, wherein the motor of the drive system is housed within the operator housing, and wherein the

sprocket and the toothed gear portion of the cap actuator are arranged external to the operator housing.

15. The remotely controlled valve assembly of claim 14, wherein: the drive system further comprises a drive stem extending from and rotationally driven by the motor; the drive stem engages a sprocket flange of the sprocket; and the sprocket flange engages a housing opening of the operator housing.

16. The remotely controlled valve assembly of claim 15, wherein a packing is mounted to the sprocket flange and is arranged within the housing opening to form a fluid-tight seal between the sprocket flange and the operator housing.

17. The remotely controlled valve assembly of claim 14, wherein: the operator housing comprises a drive housing portion defining a drive housing cavity and an actuator housing portion defining an actuator housing cavity; the motor is housed within the drive housing cavity; and at least a portion of the cap engagement portion of the cap actuator is housed within the actuator housing cavity.

18. The remotely controlled valve assembly of claim 11, wherein: the remotely controlled valve operator further comprises a torque transmission device; the torque transmission device comprising a torque bar, a first foot mounted on the torque bar proximate to a first end of the torque bar, and a second foot mounted on the torque bar proximate to an opposite second end of the torque bar; and the first foot is configured to secure to a first wall of a meter box and the second foot is configured to secure to an opposing second wall of the meter box.

19. The remotely controlled valve assembly of claim 18, wherein: an operator housing of the remotely controlled valve operator comprises a mounting bracket defining a bar opening therethrough; the torque bar is formed as a threaded rod and extends through the bar opening; a first positioning nut is mounted on the threaded rod and tightened against a first bracket side of the mounting bracket; and a second positioning nut is mounted on the threaded rod and tightened against an opposite second bracket side of the mounting bracket.

20. The remotely controlled valve assembly of claim 18, wherein: the torque bar is formed as a threaded rod; each of the first foot and the second foot defines a rod fastener, a wall bracket, and a body portion extending between the rod fastener and the wall bracket; the wall bracket of the first foot is configured to secure to the first wall of the meter box and the wall bracket of the second foot is configured to secure to the opposing second wall of the meter box; the rod fastener of each of the first foot and the second foot defines a threaded bore; and the threaded rod is rotationally engaged with the threaded bore.

21. A method of remotely controlling a valve, the method comprising: providing a remotely controlled valve operator, the remotely controlled valve operator comprising: a cap actuator comprising a cap engagement portion and a toothed gear portion, the cap engagement portion engaging a valve cap of the valve; and a drive system comprising a sprocket engaging the toothed gear portion of the cap actuator and a motor for driving rotation of the sprocket, the motor remotely controlled by a controller; remotely actuating the motor with the controller to rotate the sprocket; rotating the cap actuator between a deactivated position and an activated position with the sprocket; and actuating the valve cap with the cap actuator between a closed position, wherein fluid is prohibited from flowing through the valve, and an open position, wherein fluid is permitted to flow through the valve.